

# Public Procurement, the Nature of Innovation, and Growth: Firm-Level Evidence from Germany

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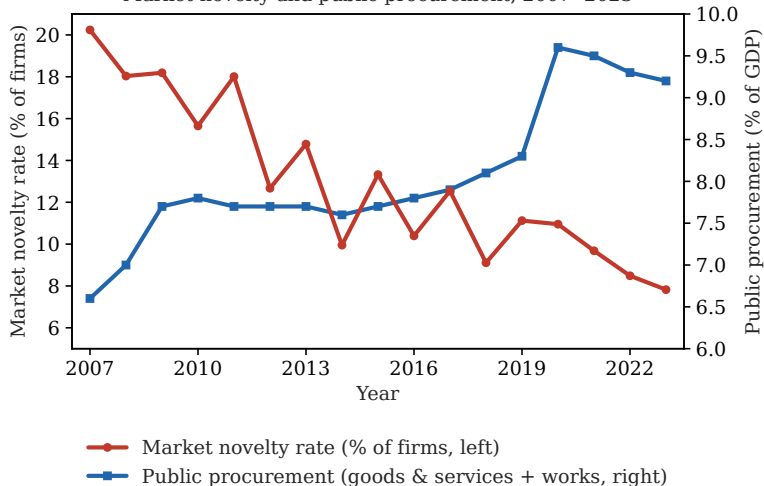
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# Public Procurement Has Risen as Market Novelty Has Fallen

Market novelty and public procurement, 2007--2023



Appendix: by industry

## Can Public Demand Shape What Firms Build?

- Government contracts come with requirements: technical standards, compatibility, certification, delivery schedules.
- Winning often depends on reliable compliance, not on having the newest idea.
- Firms in those contracts build toward requirement-compliant products, not necessarily the frontier.
- Winning expands their scale, reinforcing the shift toward existing products.

## What we do and what we find

- **Measure the innovation margin shaped by public demand**
  - Combine German firm innovation histories with public-contract award records.
  - Ask whether firms in more exposed industries report a different innovation mix.
  - Use contract cases to check the mechanism.
  - Find that higher procurement exposure is associated with **less frontier innovation** and **more incremental innovation**.
- **Quantify the growth tradeoff with an endogenous growth model**
  - Quality-ladder model where firms choose frontier and incremental innovation.
  - Estimate the model by GMM using the firm-level micro moments.
  - Disciplines scale gains against the cost of shifting innovation composition.
  - An R&D tax credit raises growth by **9.77 bp/yr** and welfare by **1.47%**.

# Outline

- 1 Data and Empirical Evidence
- 2 Model
- 3 Quantitative Analysis
- 4 Summary

## We Link Contracts to Firm Innovation

- **MIP:** firm innovation outcomes, 2009–2023 (80,028 firm-years, 23,547 firms).
- **TED:** German public contract awards by industry and year (316,642 unique contracts mapped to MIP industries).
- **ORBIS:** winner scale and productivity outcomes (126,700 firm-years, 15,493 firms).
- **BHP:** establishment entry/exit records, 2009–2023 (22.5M establishment-years, 5.07% annual exit hazard).

## Three Contracts, Read Closely

### Vaccine regulator, 2020: microscope upgrade (EUR 539K)

Detail: required patented imaging technology sold by only one manufacturer.

Story fits: the requirement rules out every other supplier by design, won by Leica.

### Agricultural research institute, 2020: new lab building (EUR 1.07M)

Detail: required licensed fire-safety and utilities engineers, provided by few firms.

Story fits: a narrow technical requirement, won by M+M.

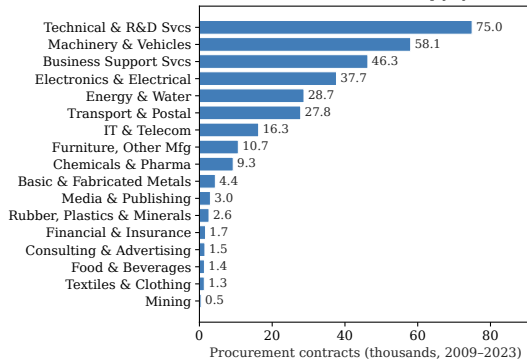
### Cybersecurity agency, 2021: AI for factory-network security (EUR 216K)

Detail: 5,000 words on risk, but universities, security firms, and research labs could all bid.

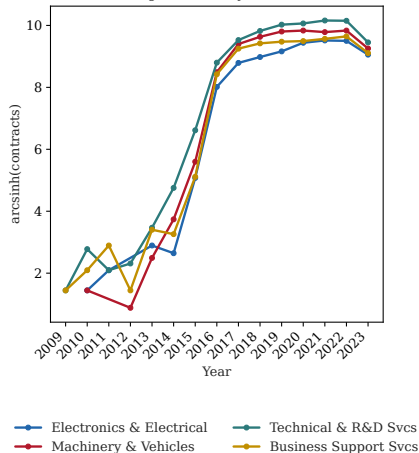
Story does not fit: detailed requirements, but a wide, competitive field, won by Fraunhofer.

# Procurement Exposure Varies Across Sectors and Time

(A) Contract volume varies sharply by sector



(B) And has grown steadily within each sector



Appendix: by industry

Which sectors are held back most



## We Compare Firms as Sector Procurement Exposure Changes

$$Y_{ist} = \alpha_i + \delta_t + \beta \operatorname{arcsinh}(\text{contracts}_{s,t}) + \varepsilon_{ist}$$

- Firm fixed effects compare each firm to itself across survey waves. Year fixed effects absorb aggregate German innovation shocks.
- $\beta$  summarizes how outcomes move with sector-year contract exposure.

| Outcome                  | Margin                                   |
|--------------------------|------------------------------------------|
| Market novelty           | New to the market, not just to the firm  |
| Existing-product revenue | Turnover share from unchanged products   |
| Cost-reduction revenue   | Turnover share from cost-cutting changes |

## Procurement Exposure Is Linked to *What* Firms Innovate

|                   | (1)<br>Mkt novelty | (2)<br>Exist.-prod. rev. | (3)<br>Cost-red. rev. |
|-------------------|--------------------|--------------------------|-----------------------|
| Proc. exposure    | -0.199*<br>(0.101) | 0.379**<br>(0.137)       | -0.047*<br>(0.024)    |
| +1 SD assoc. (pp) | -0.709             | 1.352                    | -0.162                |

Within-firm OLS, firm and year FEs; SEs (parentheses) clustered by MIP sector (21 clusters). Treatment:  $\text{arcsinh}(\text{contract count})$  in the firm's MIP sector-year, 2009–2023. Col. 2 is the turnover share from unchanged products; col. 3 is the share from cost-reducing innovations.  $N$ : 66,980 / 25,962 / 40,635. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

By type/buyer

Winner outcomes

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## **Standard Quality-Ladder Model with Endogenous Innovation Direction**

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- Consists of firms with a production and innovation choice, households, and a government that procures incremental lines.
- Firms choose how much effort to put into R&D and what share of that effort to direct toward frontier vs. incremental innovation.
- The government sets a procurement demand for incremental lines, which raises their profits and shifts the direction of innovation.
- The model is set in continuous time and solved on the steady-state.

## Firms Choose Effort and Direction

$$y_j = q_j \ell_j^P$$

$$q_j \xrightarrow{\text{rate } x} \begin{cases} \lambda_I q_j & \text{share } 1 - \mu \text{ (incremental)} \\ \lambda_F q_j & \text{share } \mu \text{ (frontier)} \end{cases} \quad \lambda_F > \lambda_I > 1$$

$$\ell_R(x, \mu) = c_x x^\zeta + \frac{\kappa}{\eta} \mu^\eta$$

- Each line  $j$  has quality  $q_j$  and production labor  $\ell_j^P$ ; a firm is a portfolio of lines.
- Effort  $x$  sets how often a line improves; direction  $\mu$  sets share on the frontier.
- $\ell_R(x, \mu)$  is the labor cost of effort and direction, labor cost  $w$ .

## Government Demand Rewards Incremental Lines

$$T = (1 - \mu^*)G$$

- Procurement is financed by a lump-sum tax.
- $G$  is government demand per incremental line.
- Procurement raises profits on requirement-compliant incremental lines:

$$\pi_I(G) = \bar{\pi}_I(1 + G), \quad \pi_F(G) = \bar{\pi}_F, \quad \bar{\pi}_k = \frac{\lambda_k - 1}{\lambda_k}$$

Why incremental only?

How much of this is real?

## Firm Value Trades Off Effort and Direction

$$\underbrace{(r + \tau)v_k}_{\text{value}} = \underbrace{\pi_k(G)}_{\text{profits}} - \underbrace{w \ell_R(x, \mu)}_{\text{innovation costs}} + \underbrace{x \mathbb{E}[v]}_{\text{new-line value}}, \quad k \in \{I, F\}$$

$$c_x \zeta x^{\zeta-1} w = \mathbb{E}[v]$$

$$w \kappa \mu^{\eta-1} = x (v_F - v_I)$$

- Effort responds to the expected value of a new line.
- Direction responds to the value premium of a frontier line.
- Free entry pins down  $\mathbb{E}[v] = wh_E$ .

## Equilibrium Definition

- A **balanced-growth equilibrium** is a set of allocations  $(v_I, v_F, x^*, \mu^*, \tau^*)$  and a wage  $w^*$  such that:
  - 1 Firms maximize value:  $(v_I, v_F, x^*, \mu^*)$  solve the Bellman equations given  $w^*$  and government demand  $G$ .
  - 2 Households maximize utility given  $w^*$  and the procurement tax  $T$ , pinning down  $r = \rho + g$ .
  - 3 Free entry holds:  $\mathbb{E}[v](\mu^*; G) = w^* h_E$ .
  - 4 Labor markets clear:  $\bar{L} = L_P + \ell_R + h_E(\tau^* - x^*)$ .
  - 5 The economy grows at a constant rate  $g = x^*[(1 - \mu^*) \log \lambda_I + \mu^* \log \lambda_F]$ .



## Procurement Changes What Firms Build

$$v_F(G) - v_I(G) = \frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{r + \tau}$$

- Higher  $G$  raises the payoff to incremental lines.
- The frontier premium narrows.
- Firms choose a lower frontier share:

$$G \uparrow \Rightarrow \mu^*(G) \downarrow$$

- With fixed labor, more production demand can also reduce R&D labor.

## Growth Falls Entirely Through Composition

$$g = x[(1 - \mu) \log \lambda_I + \mu \log \lambda_F], \quad \lambda_F > \lambda_I$$

- Free entry pins effort independently of  $G$
- Only composition responds to procurement.

$$G \uparrow \Rightarrow \mu^* \downarrow \Rightarrow g \downarrow$$

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## Strategy Overview

- Take some parameters from the literature and national accounts, and estimate the main drivers of innovation composition and growth.
- Use the estimated model to quantify the effects of procurement on growth and innovation composition.
- Analyze counterfactuals that redirect procurement support to R&D, relax requirements, or roll back procurement to historical levels.

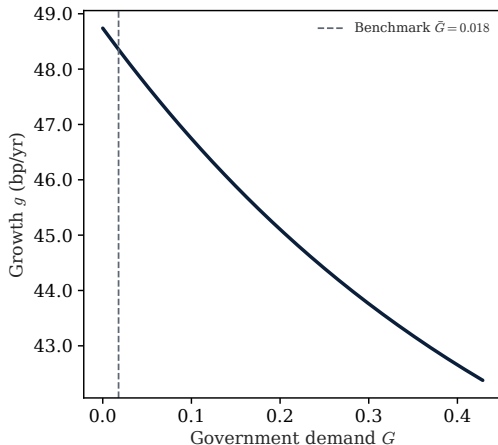
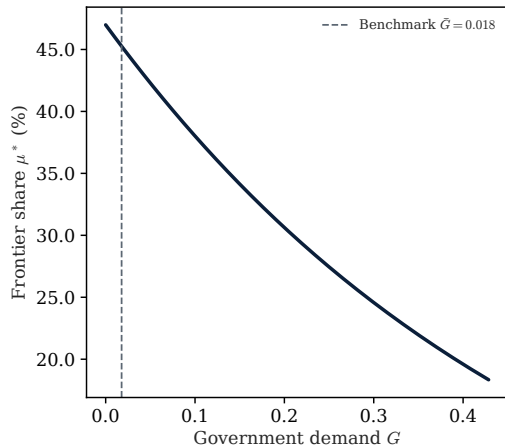
## The Model Matches Five Targeted Moments

| Moment                                   | Data | (SE)  | Model |
|------------------------------------------|------|-------|-------|
| Frontier share among innovators (%)      | 42.7 | (0.5) | 45.2  |
| Frontier revenue share, new products (%) | 58.2 | (0.5) | 55.6  |
| Large-firm share (%)                     | 10.4 | (0.3) | 10.6  |
| Creative-destruction rate (%)            | 5.1  | (0.6) | 5.7   |
| R&D personnel share (%)                  | 2.9  | (0.1) | 3.3   |

## The Estimates Put Frontier Steps Above Incremental Steps

| Parameter                                               | Estimate (SE) | Interpretation / source                 |
|---------------------------------------------------------|---------------|-----------------------------------------|
| <i>Jointly estimated (GMM, 5 moments, 4 parameters)</i> |               |                                         |
| $\lambda_F$ frontier quality step                       | 1.150 (0.003) | raises quality 15.0% per event          |
| $\lambda_I$ incremental quality step                    | 1.093 (0.003) | raises quality 9.3% per event           |
| $\kappa$ direction cost                                 | 0.044 (0.004) | cost of steering effort to the frontier |
| $h_E$ entry labor cost                                  | 1.338 (0.013) | labor per unit of creative destruction  |
| <i>Externally calibrated</i>                            |               |                                         |
| $\bar{g}$ target growth rate                            | 0.005         | rounded, within PWT ranges              |
| $\bar{G}$ benchmark government demand                   | 0.018         | Vergabestatistik / Destatis             |
| $\zeta$ effort-cost curvature                           | 2.000         | Acemoglu et al. (2018)                  |
| $\eta$ direction-cost curvature                         | 2.000         | calibrated                              |
| <i>Implied and recovered</i>                            |               |                                         |
| $w$ wage                                                | 0.953         | endogenous benchmark wage               |
| $\bar{\pi}_F$ frontier per-line profit                  | 0.130         | Bertrand markup, from $\lambda_F$       |
| $\bar{\pi}_I$ incremental per-line profit               | 0.085         | Bertrand markup, from $\lambda_I$       |

# Raising Procurement Lowers Frontier Innovation and Growth



## The Largest Gains Come From Redirecting Support to R&D

- **R&D tax credit:** redirect the procurement budget to a proportional R&D credit, sized to 2020 policy (25% of R&D costs). (Dechezleprêtre et al. 2023)
- **Requirement realignment:** keep spending fixed, but let frontier and incremental lines compete on open performance criteria. (Howell et al. 2025; Edquist & Zabala-Iturriagagoitia 2020)
- **Historical rollback:** scale government demand back to its 2008 share of GDP (1.3% vs. 1.8% now). (OECD national accounts)

| Scenario                | $\Delta\mu^*$ | $\Delta\ell_R$ (%) | $\Delta g$ (bp/yr) | CEV (%) |
|-------------------------|---------------|--------------------|--------------------|---------|
| R&D tax credit          | 0.071         | +18.74             | +9.77              | 1.47    |
| Requirement realignment | 0.018         | 1.05               | 0.39               | 0.09    |
| Historical rollback     | 0.004         | 0.26               | 0.10               | 0.023   |



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## Procurement Policy Has an Innovation-Composition Tradeoff

- German firm microdata show that higher procurement exposure is associated with less frontier innovation and more incremental innovation.
- A standard quality-ladder model with endogenous innovation direction and effort rationalizes the mechanism and quantifies the growth tradeoff.
- We estimate the model using the same microdata as moments and discipline the scale of government demand with national accounts.
- The counterfactuals show that redirecting procurement support to R&D raises growth and welfare more than requirement realignment or historical rollbacks.

Thanks! You can send me any comments or suggestions at  
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## Outline

### 5 Appendix

## Where We Sit in the Literature

- **Innovation composition and growth:** (Garcia-Macia, Hsieh & Klenow 2019; Acemoglu, Akcigit & Celik 2022). We add a procurement-demand margin that shifts the direction of firm innovation.
- **Demand-side innovation policy:** (Krieger, Prüfer & Strecke 2024; Czarnitzki, Hünermund & Moshgbar 2020). Procurement redirects composition, distinct from subsidies that raise R&D levels.
- **Procurement contract design:** (Howell, Rathje, Van Reenen & Wong 2025; Clemens & Rogers 2026). We quantify the same design margin in general equilibrium.

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## Appendix: Winners Scale Up Without Clear Productivity Gains

|                      | (1)<br>Employment   | (2)<br>Turnover     | (3)<br>Labor prod. | (4)<br>Return on capital |
|----------------------|---------------------|---------------------|--------------------|--------------------------|
| Contracts won        | 0.025***<br>(0.002) | 0.029***<br>(0.007) | 0.007<br>(0.007)   | 0.001<br>(0.001)         |
| Effect of +1 SD (pp) | 0.023               | 0.028               | 0.006              | 0.001                    |

Within-firm OLS on  $\arcsinh(\text{contracts won})$ , firm and year FEs; SEs clustered by NACE section.  
Sample: ORBIS winner panel, 2009–2023, 15,493 matched firms.  $N$ : 88,706 / 38,690 / 33,994 / 28,584.

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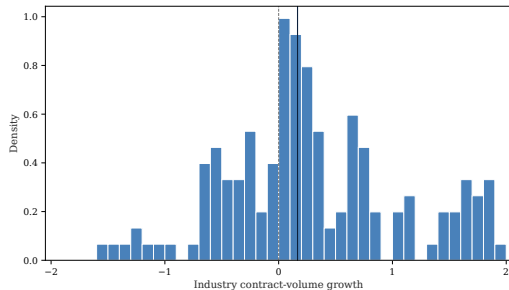
## The Association Varies by Purchase Type and Buyer

|                     | (1) Mkt novelty     | (2) Exist.-prod. rev. | (3) Cost-red. rev. |
|---------------------|---------------------|-----------------------|--------------------|
| Supplies & services | -0.211**<br>(0.099) | 0.393***<br>(0.137)   | -0.046*<br>(0.023) |
| Works               | -0.010<br>(0.142)   | 0.180<br>(0.120)      | 0.024<br>(0.022)   |
| Central government  | -0.291**<br>(0.131) | 0.518**<br>(0.194)    | -0.045<br>(0.037)  |

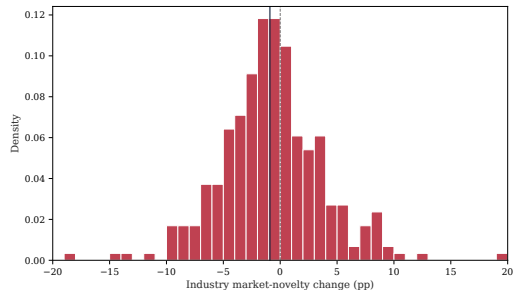
Each row is a separate within-firm OLS regression, firm and year FEs; SEs clustered by MIP sector.  
Treatment:  $\text{arcsinh}(\text{sector-year contract count})$  for the indicated type or buyer.  $N$ : 66,980 / 25,962 / 40,635.

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## Appendix: Sector Variation Behind the Motivation



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## Appendix: Why Does Procurement Favor Incremental Lines?

- Frontier lines promise higher expected quality but higher delivery risk: harder to specify, more likely to slip or fail.
- Officers face career penalties for failed or delayed projects, not for missed upside, and act like risk-averse buyers. (Prendergast 2007; Bandiera, Prat & Valletti 2009)
- Microfoundation: government mean-variance preferences over delivered quality, risk aversion  $\gamma_G \geq 0$ , rationalize  $G_F = 0$  above a threshold; not separately estimated, since our records show contract design, not delivery risk.
- Risk aversion is one candidate mechanism. The model only needs a demand wedge; legal scoring rules and audit requirements could generate the same asymmetry.

## Appendix: Where Procurement Holds Back the Frontier Most

| Sector                | Frontier-share gain if procurement removed (pp) |
|-----------------------|-------------------------------------------------|
| Manufacturing         | +1.44                                           |
| Professional services | +1.35                                           |
| Transport             | +1.16                                           |
| IT services           | +1.11                                           |
| Finance               | +0.67                                           |
| Mining                | +0.48                                           |

- Sector wedges  $G_s$  mapped from mean sector-year contract counts, normalized to the benchmark  $\bar{G} = 0.018$ .
- Procurement holds back the frontier most exactly where it buys the most.

## How Much of the Wedge Is Real?

- No data identify one preferred frontier-budget share  $\alpha_F$  directly. Contract design gives a capacity proxy: score-based (MEAT) awards can weigh quality beyond price, price-only awards cannot.
- 50.6% of contract value is score-based, close to  $\alpha_F = 0.50$ : a capacity proxy, not a causal estimate, since MEAT exposure alone does not raise market novelty in the reduced-form split.

| $\alpha_F$        | $\Delta\mu^*$ (pp) | $\Delta g$ (bp/yr) | CEV (%) |
|-------------------|--------------------|--------------------|---------|
| 0.10              | 0.38               | 0.08               | 0.014   |
| 0.25              | 0.94               | 0.21               | 0.045   |
| 0.50 <sup>†</sup> | 1.86               | 0.41               | 0.096   |
| 0.75              | 2.77               | 0.61               | 0.148   |

<sup>†</sup> Lands almost exactly on the equal-treatment case in the main text ( $\alpha_F \approx \mu^* \approx 0.47$ ).